# Project Initialization and Planning Phase
# Project Proposal (Proposed Solution)

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| **Date** | 31 January 2026 |
| **Team ID** | anshrathore0202-cyber |
| **Project Title** | F1 Racing Intelligence Analysis |
| **Maximum Marks** | 3 Marks |

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and data specifications.

| **Project Overview** | |
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| **Objective** | Develop an end-to-end Formula 1 Business Intelligence and data analytics pipeline that transforms multi-season raw telemetry, driver standings, and race results into interactive visual dashboards for deep performance analysis. |
| **Scope** | Covers automated data extraction, cleaning, schema transformation, KPI computation (pit efficiency, stint lap pace, qualifying-to-finish position delta), and interactive Tableau dashboard creation for historical F1 seasons. |
| **Problem Statement** | |
| **Description** | Formula 1 race data is highly fragmented across raw CSV files, static web tables, and unstandardized API endpoints, making multi-season trend exploration and head-to-head driver analysis complex and time-consuming. |
| **Impact** | Solving this problem enables analysts, team strategists, and fans to instantly uncover hidden driver performance trends, track-specific strategy impacts, and constructor dominance metrics through clean, interactive visuals. |
| **Proposed Solution** | |
| **Approach** | Implement a robust ETL pipeline in Python to clean and aggregate raw Ergast F1 telemetry datasets, followed by building interactive analytical dashboards in Tableau Desktop for dynamic user exploration and story-driven insights. |
| **Key Features** | • Automated batch cleaning & relational database structuring.<br>• Interactive Tableau dashboards with dynamic season, driver, team, and circuit filters.<br>• Head-to-head lap pace overlays, stint degradation heatmaps, and pit-stop efficiency tracking. |

| **Resource Requirements** | | |
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| **Resource Type** | **Description** | **Specification / Allocation** |
| **Hardware** | Computing Resources (CPU/GPU specifications) | Intel Core i5 / AMD Ryzen 5 (8 Cores) |
| | Memory (RAM specifications) | 16 GB DDR4 RAM |
| | Storage (Disk space for data and logs) | 512 GB NVMe SSD Storage |

| Resource Requirements | | |
|---|---|---|
| **Resource Type** | **Description** | **Specification / Allocation** |
| **Software** | Frameworks & Languages | Python 3.10+, SQL, HTML5/CSS3 |
| | Libraries & Tools | Pandas, NumPy, Matplotlib, Seaborn |
| | Development & BI Environment | VS Code, Tableau Desktop / Public, Git |
| **Data** | Data Source, Size, and Format | Ergast F1 Database / Kaggle CSV files (~100,000+ telemetry rows) |